# EXPERIMENT #7

Timestamp: 20260317_094516 | Environment: Local

## 1. Experiment Parameters & Network Statistics

| Parameter | Value |
|---|---|
| Number of sticks (N) | 30 |
| Stick length (a) | 0.7 |
| Domain size | 1.0 × 1.0 |
| Source region | x < 0.1 |
| Drain region | x > 0.9 |
| Total intersections | 61 |
| Has percolation | ✓ Yes |
| # Percolating paths | 1 |
| Nodes before pruning | 61 |
| Nodes after pruning | 18 |
| Edges before pruning | 60 |
| Edges after pruning | 17 |

## 2. Network Visualizations

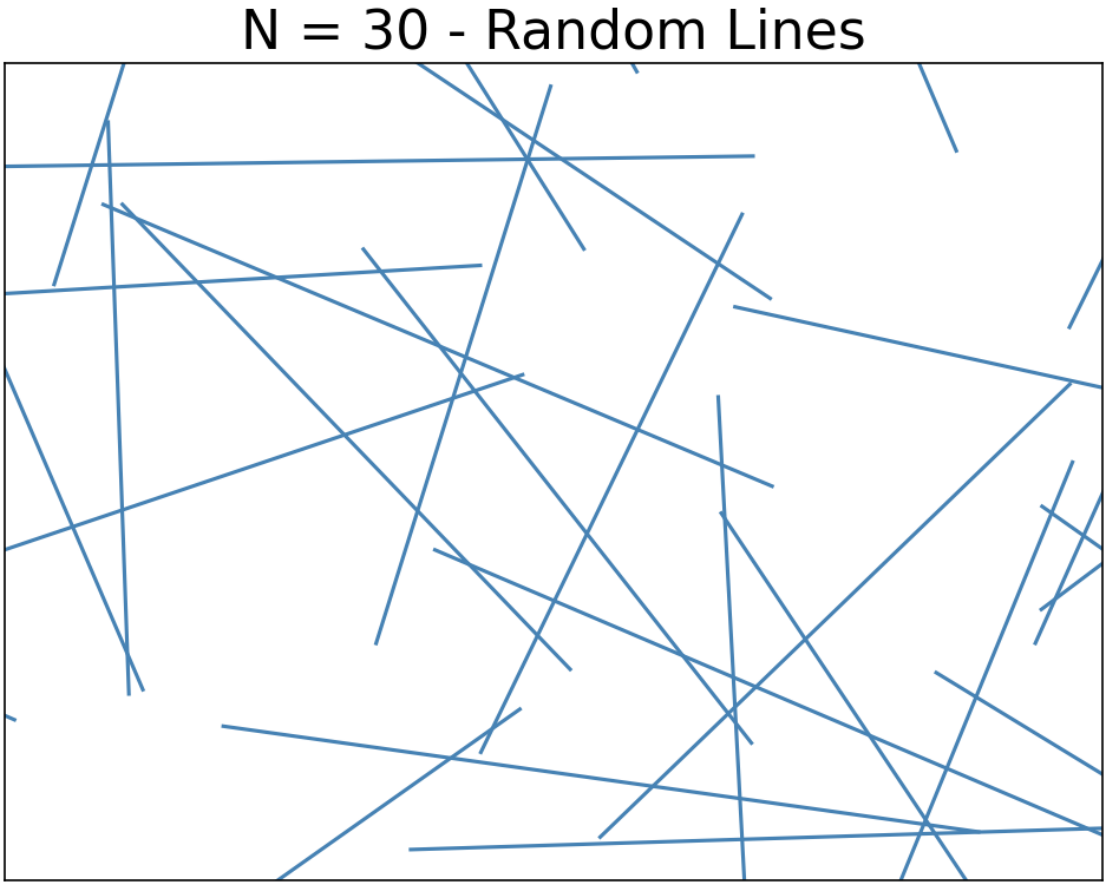



Fig. — 01 random lines N30

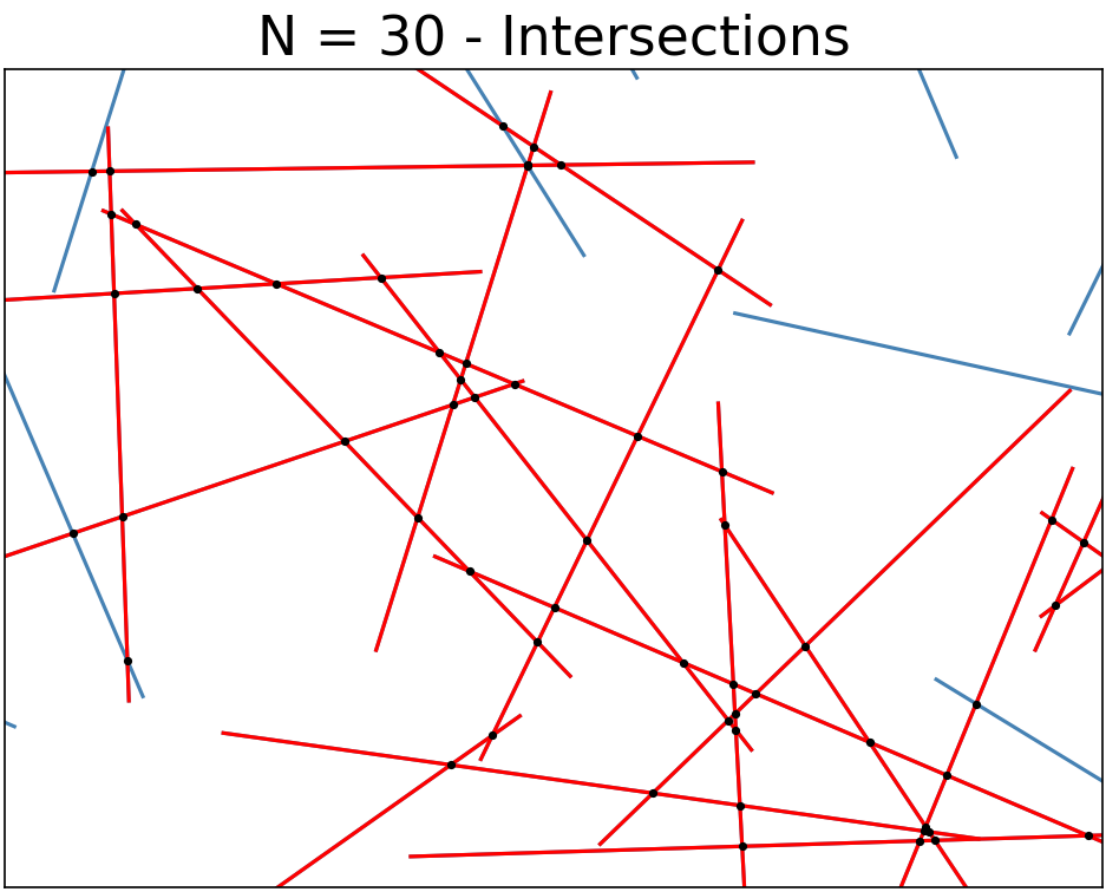



Fig. — 02 intersections N30

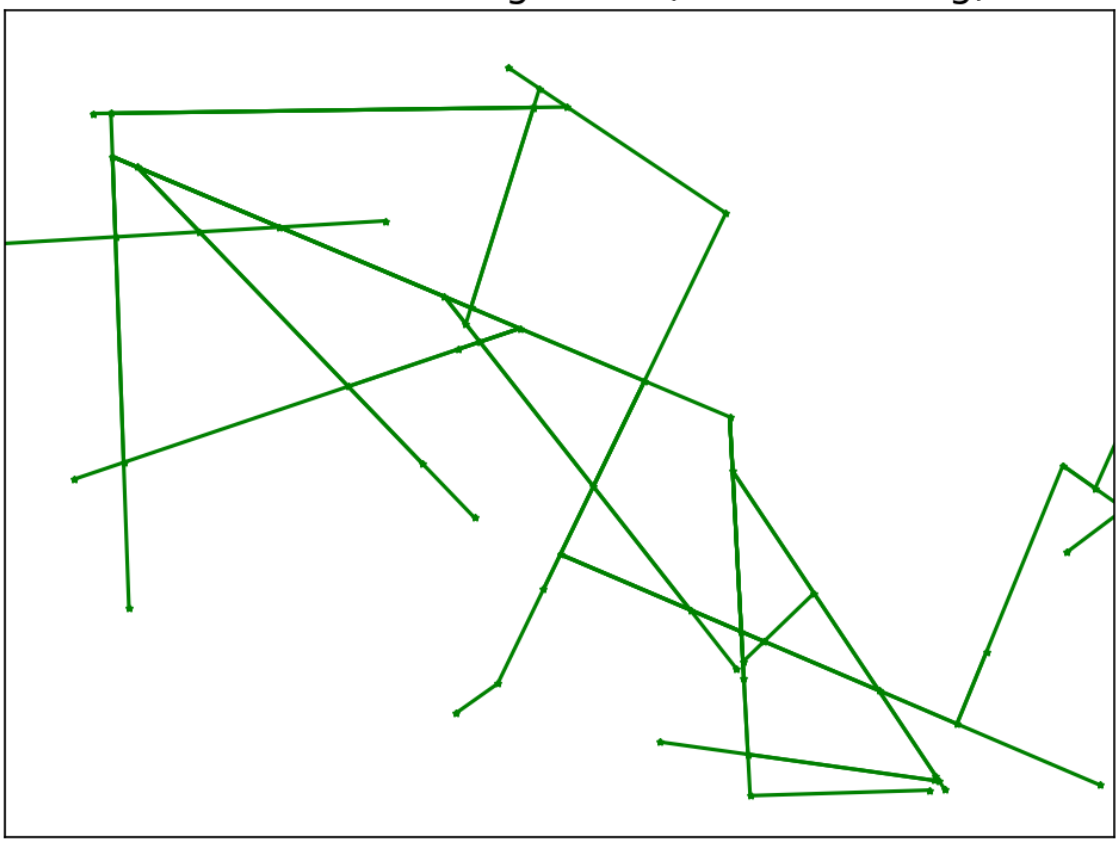



Fig. — 03 percolating paths before pruning N30

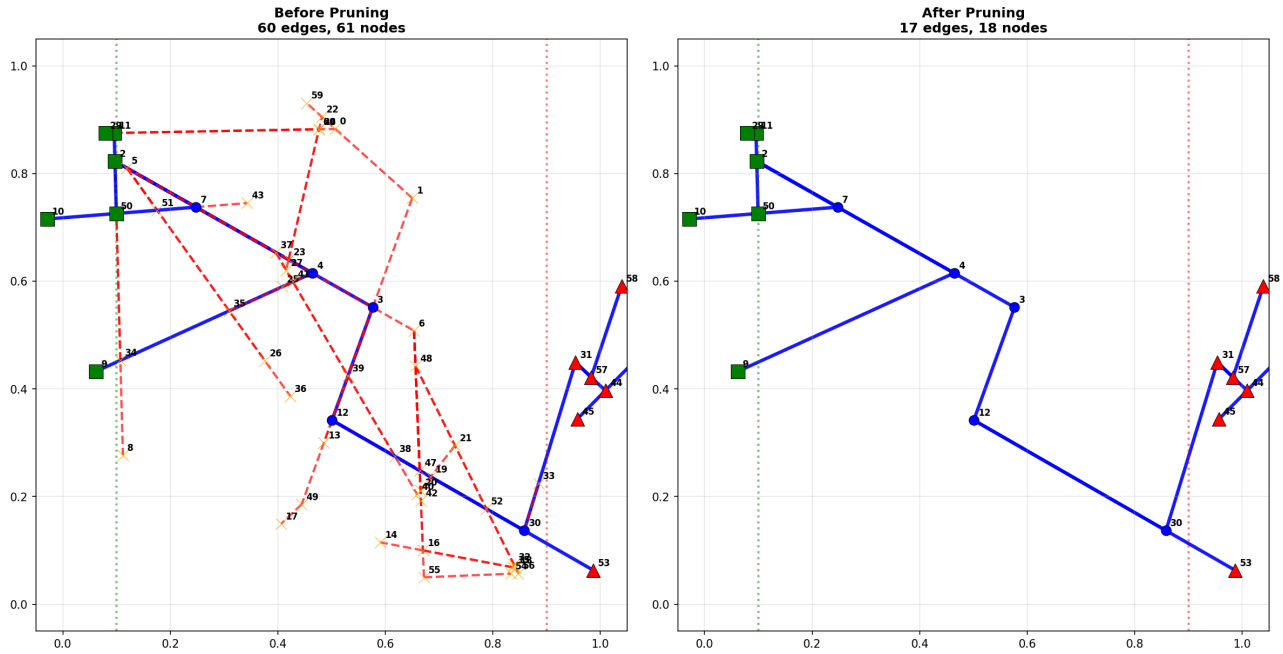



Fig. — 04 before after pruning N30

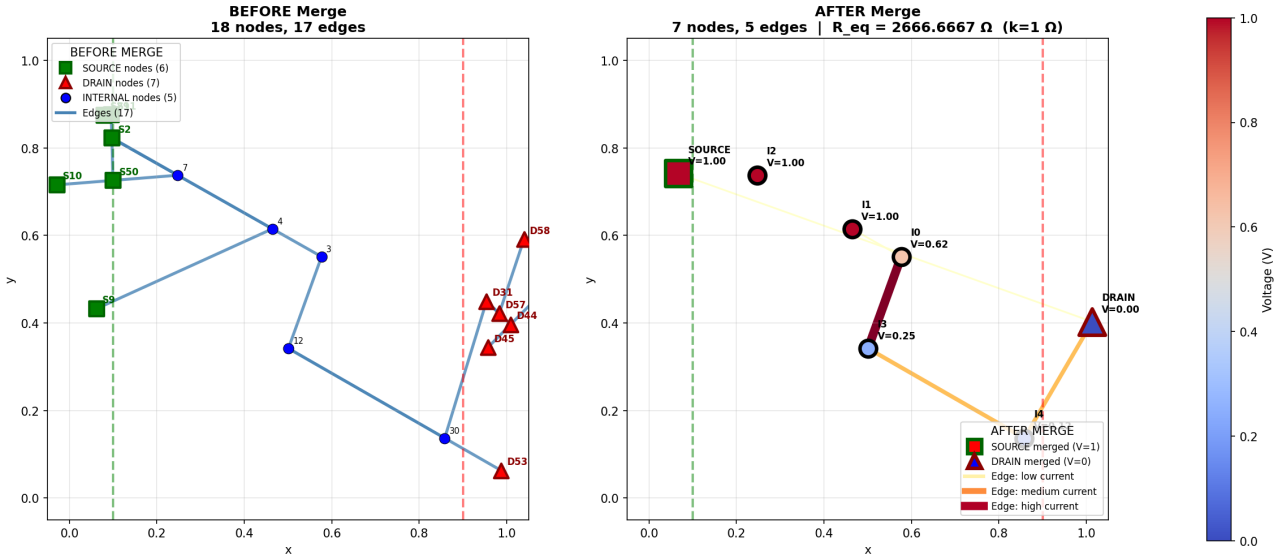



Fig. — 05 before after merge N30

## 3. Edge Resistances (Pruned Network)

| Edge # | Node 1 | Node 2 | Conductance (S) | Resistance (Ω) | Voltage Drop (V) |
|---|---|---|---|---|---|
| 0 | 2 | 3 | 1.0000 | 1.0000 | 0.750000 |
| 1 | 2 | 4 | 1.0000 | 1.0000 | 0.875000 |
| 2 | 2 | 7 | 1.0000 | 1.0000 | 1.000000 |
| 3 | 2 | 11 | 1.0000 | 1.0000 | 1.000000 |
| 4 | 2 | 50 | 1.0000 | 1.0000 | 1.000000 |
| 5 | 3 | 12 | 1.0000 | 1.0000 | 0.250000 |
| 6 | 4 | 9 | 1.0000 | 1.0000 | 0.125000 |
| 7 | 7 | 10 | 1.0000 | 1.0000 | 0.000000 |
| 8 | 11 | 29 | 1.0000 | 1.0000 | 0.000000 |
| 9 | 12 | 30 | 1.0000 | 1.0000 | 0.000000 |
| 10 | 12 | 53 | 1.0000 | 1.0000 | 0.000000 |
| 11 | 30 | 31 | 1.0000 | 1.0000 | 0.000000 |
| 12 | 31 | 44 | 1.0000 | 1.0000 | 0.000000 |
| 13 | 31 | 57 | 1.0000 | 1.0000 | 0.000000 |
| 14 | 44 | 45 | 1.0000 | 1.0000 | 0.000000 |
| 15 | 44 | 46 | 1.0000 | 1.0000 | 0.000000 |
| 16 | 57 | 58 | 1.0000 | 1.0000 | 0.000000 |

## 4. Impedance Calculation

| Quantity | Value |
|---|---|
| Applied voltage (V) | 1.00 V |
| Resistance per segment k | 1 Ω |

| | |
|---|---|
| Calculated impedance R_eq | 2666.666667 Ω |
| Total current I | 1.250000e-04 A |

## Impedance for different k (resistance per segment) values

| k (Ω/segment) | R_equivalent (Ω) |
|---|---|
| 1 | 2.666667 |
| 10 | 26.666667 |
| 100 | 266.666667 |
| 1000 | 2666.666667 |
| 10000 | 26666.666667 |